Technical Deliverable

Client 1: Technical Codebook

Refreshed PSID crisis-module workflow, scoring logic, and artifact schema

Document status

Prepared for client review
Updated: April 2026
Repository: Team-PSID

Contents

1 Purpose and Scope

This technical codebook documents the refreshed PSID crisis-module workflow from the authoritative ranked source file through the final scored CSV, figures, spreadsheet, and PDF bundle. It is written to support analytic review, reproducibility, and downstream client sign-off.

Production status

- 52 ranked historical questions in the current corpus.
- 28 selected rows in the scored module.
- 3 `Generic` rows and 49 `Specific` rows after binary thresholding.
- 29.98 selected minutes across the scored module.
- 2 demographic traceability rows retained in the master review file but excluded from the deployable instrument.

2 Headline Benchmark Summary

Metric	Value
Total ranked questions	52
Selected questions	28
Selected minutes	29.98
Full corpus minutes	68.60
Generic rows	3
Specific rows	49
Average R_i	2.201
Maximum R_i	5.667
Average P_i	2.561
Maximum P_i	7.860

Table 1: Current benchmark summary used by the maintained client package.

3 Figures and Visual Diagnostics

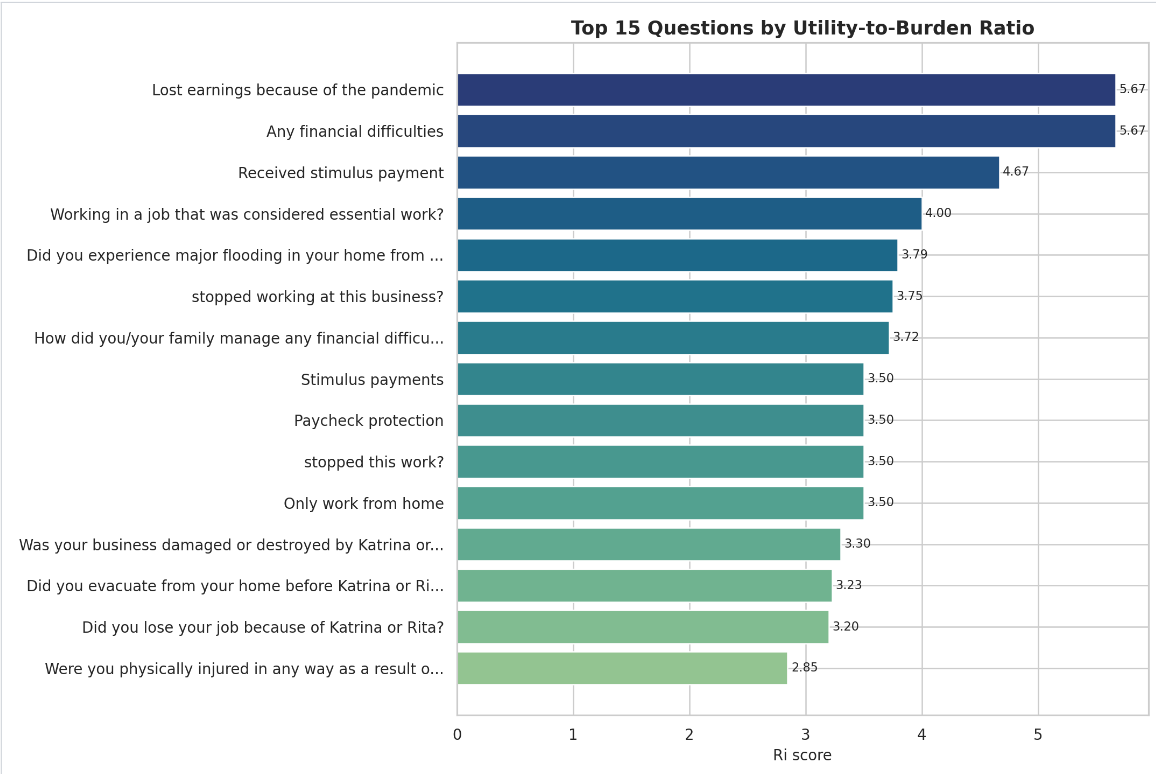


Figure 1: Top-ranked questions from the refreshed corpus.

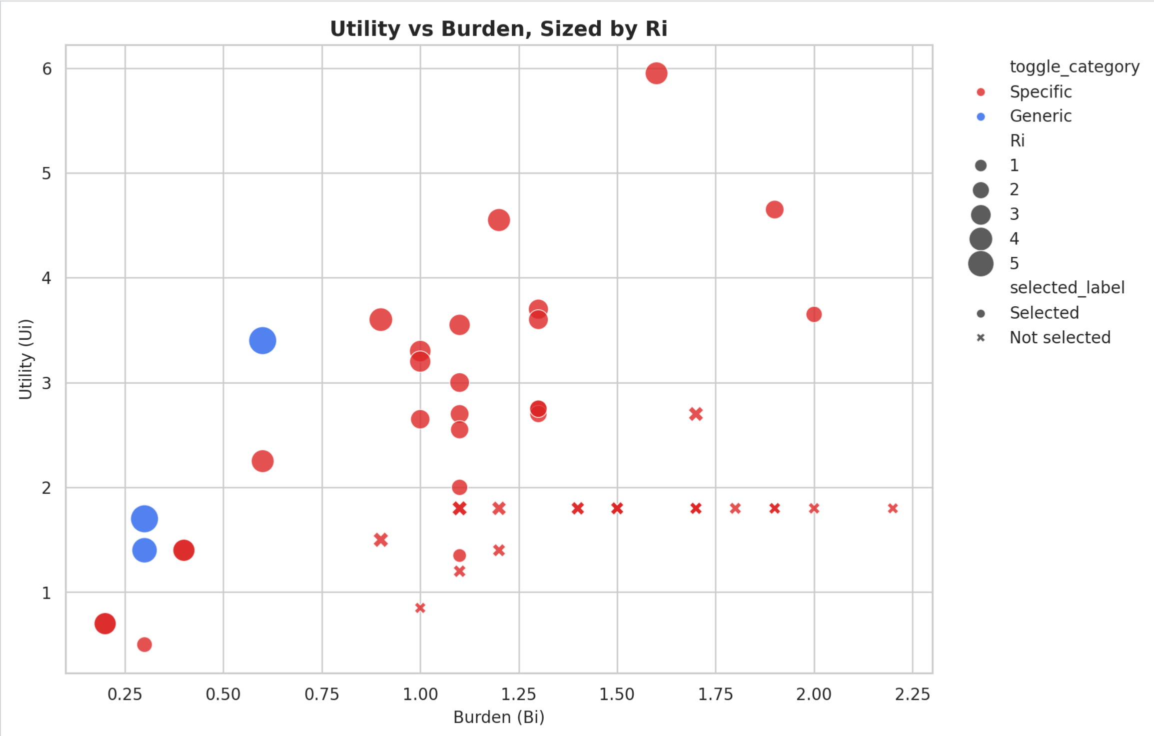
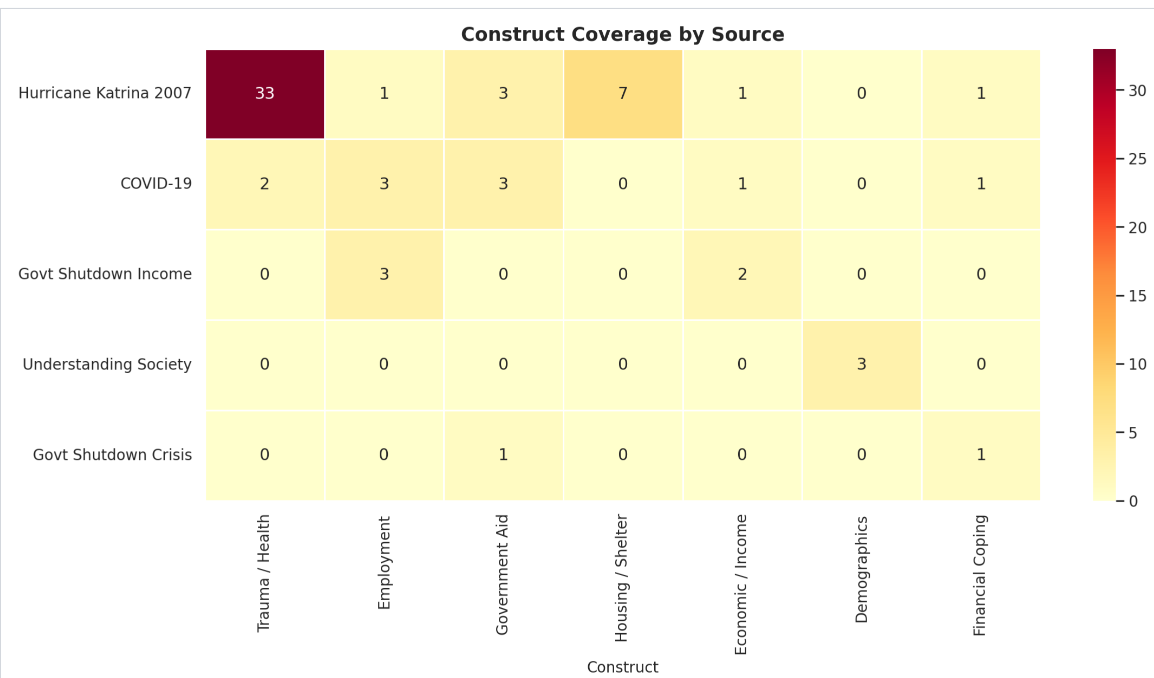


Figure 2: Utility-versus-burden view used to inspect signal efficiency.



	Trauma / Health	Employment	Government Aid	Housing / Shelter	Economic / Income	Demographics	Financial Coping
Hurricane Katrina 2007	33	1	3	7	1	0	1
COVID-19	2	3	3	0	1	0	1
Govt Shutdown Income	0	3	0	0	2	0	0
Understanding Society	0	0	0	0	0	3	0
Govt Shutdown Crisis	0	0	1	0	0	0	1

Figure 3: Construct coverage heatmap for the refreshed item bank.

4 End-to-End Workflow Pipeline

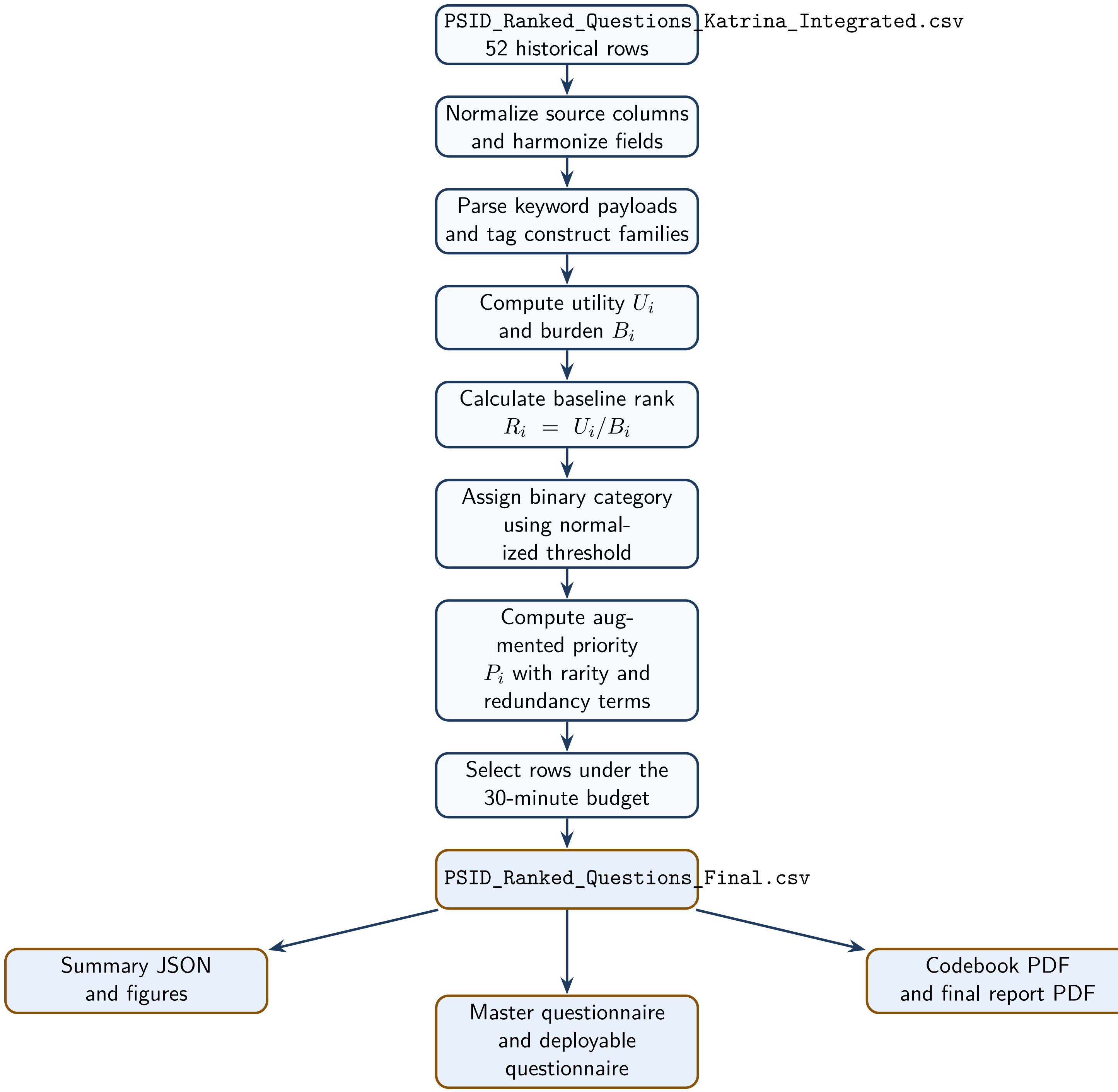


Figure 4: Production workflow from the raw integrated source to the maintained client package.

5 Files and Roles

File	Role	Why it matters
`data/PSID_Ranked_Questions_Katrina_Integrated.csv`	Raw integrated source bank	Starting point for every production rebuild.
`PSID_NLP_Crisis_Module_Structure.py`	Shared scoring helpers	Holds constants, keyword parsing, burden logic, threshold helpers, and selection utilities.
`generate_psid_artifacts.py`	Production rebuild engine	Writes the final scored CSV, summary JSON, dashboard payload, and figures.
`data/PSID_Ranked_Questions_Final.csv`	Authoritative scored output	Main ranked table used by the notebook and client deliverables.
`PSID_NLP_Crisis_Module_Final.ipynb`	Validation notebook	Recomputes the workflow and verifies the persisted artifact bundle.
`generate_client_documents.py`	Client-package renderer	Builds the spreadsheet and PDFs from the scored CSV and summary JSON.
`data/psid_artifact_summary.json`	Metric bundle	Stores benchmark counts and score summaries for the latest run.

Table 2: Files that drive the maintained workflow and client package.

6 Core Scoring Logic

6.1 Utility, burden, and baseline rank

The baseline score is driven by question utility relative to response burden.

Core formulae

$$U_i = \sum \text{keyword weights}$$
$$B_i = \max(0.10 \times \text{word\_count} + 0.20 \times \text{complexity}, 0.1)$$
$$R_i = \frac{U_i}{B_i}$$

- U_i rewards crisis-relevant content signaled by tagged keywords.
- B_i penalizes long or complex wording.
- R_i ranks items that carry more signal per unit of burden.

6.2 Binary category assignment

The client requirement was a binary rule based on R_i, but the raw score is not naturally bounded in $[0, 1]$. The production pipeline therefore applies the cutoff to a min-max normalized score.

Operational threshold rule

$$\text{ri\_threshold\_score} = \frac{R_i - \min(R_i)}{\max(R_i) - \min(R_i)}$$
$$\text{Generic} \iff \text{ri\_threshold\_score} \geq 0.70$$
$$\text{Specific} \iff \text{ri\_threshold\_score} < 0.70$$

6.3 Augmented priority score

The final prioritization score preserves R_i while rewarding breadth and distinctiveness.

Priority adjustment

$$\text{augmented\_utility} = U_i \times \Big(1 + 0.32 \times \text{idf\_scaled} + 0.24 \times \text{construct\_scaled} + 0.12 \times \text{richness\_scaled} + \text{portability\_bonus}\Big)$$

$$P_i = \frac{\text{augmented\_utility}}{B_i \times (1 + 0.65 \times \text{redundancy\_scaled})}$$

- P_i helps separate distinct, portable items from repetitive variants.
- The shortlist is still bounded by the time budget, not only by raw rank.

7 Worked Calculation Example

Example question: `GEN_LOSE_EARNINGS_PANDEMIC`

Recommended wording: *Did you lose earnings because of the pandemic?*

Field	Value	Calculation
U_i	3.400	$0.80 + 0.80 + 0.90 + 0.90$
word count	6	Counted from deployable wording
complexity	0.0	Precomputed complexity field
B_i	0.600	$\max(0.10 \times 6 + 0.20 \times 0.0, 0.1)$
R_i	5.667	$3.40/0.60$
threshold score	1.000	$(5.667 - 0.818)/(5.667 - 0.818)$
Final category	Generic	$1.000 \geq 0.70$
P_i	7.860	Enhanced priority after rarity, construct, richness, and redundancy adjustments
minutes	0.70	$(6 \times 7)/60$

Table 3: Worked example for one of the highest-value Generic items.

8 Threshold Diagnostics

Threshold diagnostic	Value
Minimum raw R_i	0.818
Maximum raw R_i	5.667
Rows with raw $R_i \geq 0.70$	52 of 52
Rows with normalized threshold score ≥ 0.70	3 of 52

Table 4: Why normalized thresholding is required for a meaningful binary split.

Interpretation

Raw $R_i \geq 0.70$ would classify every row as `Generic`. Normalized thresholding preserves the requested cutoff while still creating a real binary split between portable high-signal items and context-dependent items.

9 Construct Coverage and Interaction Notes

9.1 Selected construct coverage

Construct	Selected questions
Trauma / Health	15
Employment	7
Housing / Shelter	6
Government Aid	5
Economic / Income	4
Financial Coping	2
Demographics	2

Table 5: Construct coverage in the selected scored module.

9.2 Demographic interaction summary

Group	Rows	Average R_i	Average P_i
Demographics only	3	1.354	2.189
Non-demographic crisis items	49	2.253	2.584

Table 6: Demographic-only rows versus direct crisis-impact rows.

Interpretation

Demographic-only items remain useful for traceability and module framing, but they carry less direct crisis signal than employment, housing, aid, trauma, and economic disruption items. That is why the final deployable questionnaire excludes demographics even though the master review file retains them.

10 Output Schema

Field	Meaning
`variable_name`	Stable identifier used across the CSV, spreadsheet, and PDFs.
`question_text`	Historical source wording.
`recommended_wording`	Clean deployable wording.
`source`	Historical module source.
`toggle_category`	Final binary category: `Generic` or `Specific`.
`historical_toggle_category`	Legacy category kept for traceability.
`U_i`, `B_i`, `R_i`	Utility, burden, and baseline rank.
`ri_threshold_score`	Normalized rank used for binary categorization.
`P_i`	Enhanced priority score.
`selected_for_module`	Final selection flag under the time cap.
`minutes`	Estimated administration time.
`constructs`	Tagged construct families.

Table 7: Key fields preserved in the authoritative scored dataset.

11 Deliverables Produced From the Same Dataset

Deliverable	Audience	Purpose
`deliverables/Master_Questionnaire.xlsx`	Internal review	Full 52-row review sheet with categories, wording, and scores.
`deliverables/Deployable_Questionnaire.pdf`	Survey deployment	Final question text without demographic rows.
`deliverables/Codebook.pdf`	Technical review	Rendered technical codebook.
`deliverables/Final_Report.pdf`	Stakeholder review	Narrative summary of results and interpretation.
`deliverables/Client_1_Technical_Codebook.pdf`	Client circulation	LaTeX report version of this technical codebook.

Table 8: Client-facing and internal deliverables generated from the same scored dataset.

12 Refresh Procedure

Use this order whenever the client package is refreshed:

1. Rebuild scored artifacts via `build_all()` in `generate_psid_artifacts.py`.
2. Validate notebook outputs in `PSID_NLP_Crisis_Module_Final.ipynb`.
3. Regenerate spreadsheet and PDFs through `generate_client_documents.py`.
4. Rebuild the client-facing LaTeX reports through `docs/latex/build_client_reports.sh`.
5. Confirm that metrics, figures, tables, and deliverable paths remain aligned.

That sequence keeps the scored CSV, the notebook, the figures, the packaged PDFs, and the presentation-ready report files synchronized.